TEMPO: Physics-Informed Urban Flow Simulation via Velocity-Driven Latent Flow Matching

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Mass error vs. Latent FM
−55.0%
0.0253 → 0.0114
lower is better; same latent space, no physics

MAE vs. Latent FM
−17.0%
0.0112 → 0.0093
lowest MAE, RMSE and CRPS of all five methods

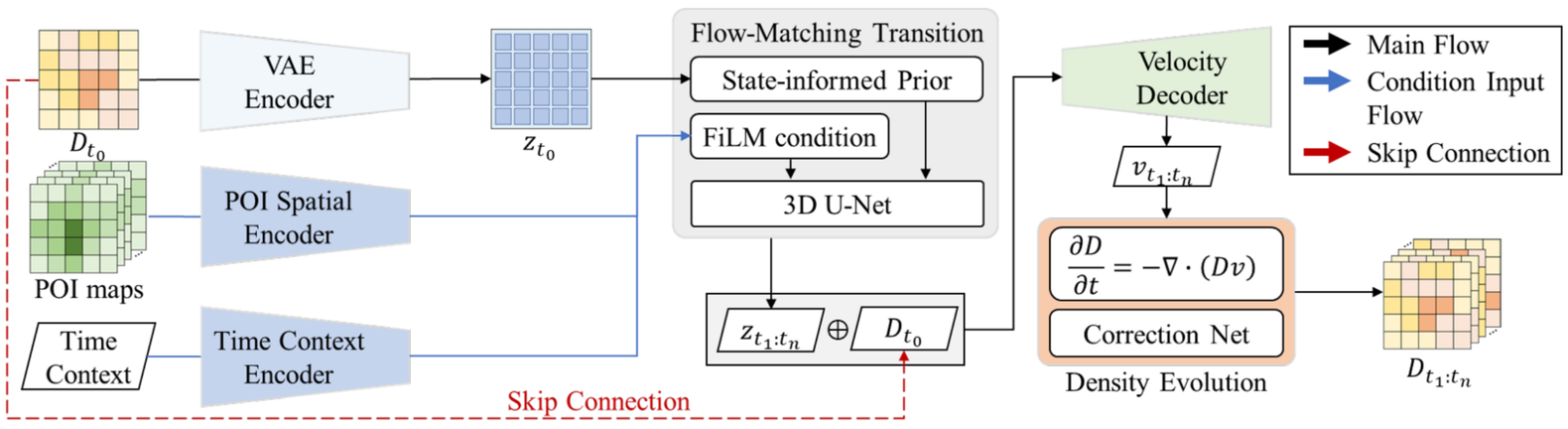
Mass error vs. Gaussian-prior TEMPO
−16.2%
0.0136 → 0.0114
ablation of the State-Informed Prior; MAE unchanged

1 Background & Motivation

- Crowd-flow simulation feeds capacity and evacuation planning; invented or lost people corrupt every count-based decision
- Deterministic predictors (ST-ResNet, Graph WaveNet): point estimates, no uncertainty
- Diffusion generators (DiffSTG, KSTDiff): no continuity in the density update, and a soft physics loss does not survive sampling noise ⇒ mass appears or vanishes

Key insight: real flow obeys **mass conservation**: a cell's density changes only through velocity-mediated transport from its neighbours. Build it in as an **architectural** inductive bias, not a loss penalty.

2 TEMPO Architecture: Velocity-Driven Latent Flow Matching



Task: given the current density map D_{t_0}, POI maps and time of day/week, generate the next T=6 density maps with uncertainty. A frozen VAE encoder maps D_{t_0} and POI maps to a latent; latent Flow Matching generates the trajectory from a **State-Informed Prior** anchored at D_{t_0}; a causal Velocity Decoder yields V; **Physics-Guided Density Evolution** advects density through a hard-coded continuity equation plus a lightweight CorrectionNet. Shaded blocks: our two contributions.

3 Method: Two Inductive Biases

1. State-Informed Prior. Standard FM starts from $\mathcal{N}(0, I)$, far from the data. We start from the encoded initial state instead:

$$x_0 \sim \mathcal{N}(z_{t_0}, \sigma^2 I), \quad \sigma{=}1; \qquad x_t = (1-t)\,x_0 + t\,x_1$$

⇒ shorter transport from the observed state. Trained with the latent OT-CFM objective [1, 2] (3D U-Net v_θ; POI via channel concat + FiLM).

2. Hard-coded continuity equation (velocity V_t is its own state, cf. second-order crowd models [6]):

$$D_{t+1} = \mathrm{ReLU}\Big(D_t - \Delta t\,\nabla\cdot(D_t V_t) + \mathrm{Corr}_\psi(D_t, V_t)\Big)$$

ReLU keeps density ≥ 0; rolled out $T = 6$ steps.

Parameter-free physics core: the divergence uses fixed 3×3 central-difference kernels (non-trainable buffers, no learnable physics parameters). The only learned part of the density update is the lightweight CorrectionNet.

Training, three decoupled stages: VAE (ELBO, then frozen) → Flow Matching U-Net → fine-tune Velocity Decoder + CorrectionNet on **FM-sampled** latents, which removes the train–inference shift at the decoder input.

4 Main Results on YJMob100K

Setup: YJMob100K, 100K individuals, 75 days, 64×64 grid, 30-min steps; input: one initial density map, 10 POI classes, time of day/week; horizon $T = 6$ (3 h); log-normalised density; mean over 5 seeds.

Method	MAE↓	RMSE↓	Mass↓	CRPS↓
ST-ResNet	0.0200	0.0335	0.0293	–
Graph WaveNet	0.0496	0.0648	0.1756	–
DiffSTG	0.0225	0.0354	0.1560	0.0196
Latent FM	0.0112	0.0169	0.0253	0.0081
TEMPO	**0.0093**	**0.0152**	**0.0114**	**0.0076**

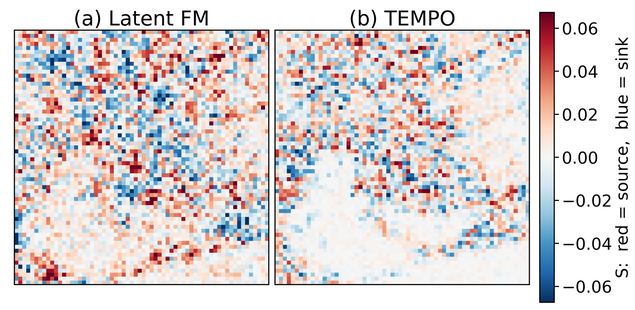
Mass = mean relative error of total predicted vs. ground-truth density per step (log-normalised space).

vs. Latent FM (same latent space): mass error 0.0253 → 0.0114, **−55.0%**, with better MAE, RMSE and CRPS.

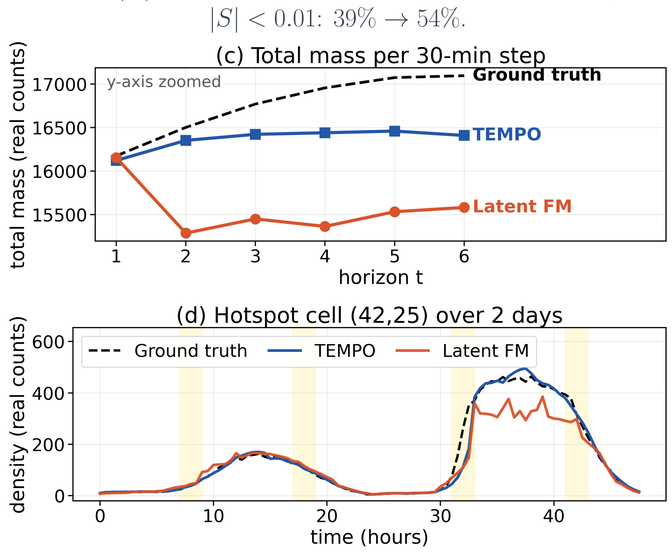
vs. DiffSTG: mass error 0.1560 ⇒ black-box diffusion decouples statistics from physics.

Trade-off: narrower predictive intervals (Cal-50% 0.42 vs. ideal 0.50; Latent FM 0.53) and lower sample diversity (0.0071 vs. 0.0159); the Gaussian-prior variant has the same diversity but higher mass error.

5 Physical Plausibility vs. Latent FM



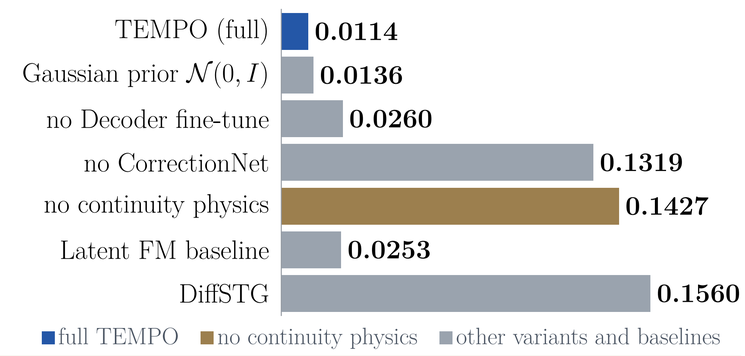
Continuity residuals (same mass-conserving optical flow), Latent FM → TEMPO: mean $|S|$ 0.019 → 0.014; net source ΣS +8.1 → −0.1; cells with $|S| < 0.01$: 39% → 54%.



Top: total mass per step, mean error 7.1% → 2.3%. Bottom: one hotspot over 2 days, 16 consecutive 3-h windows from the observed state (8-sample mean): TEMPO tracks the commuting peaks, Latent FM undershoots them.

All three physical signatures improve.

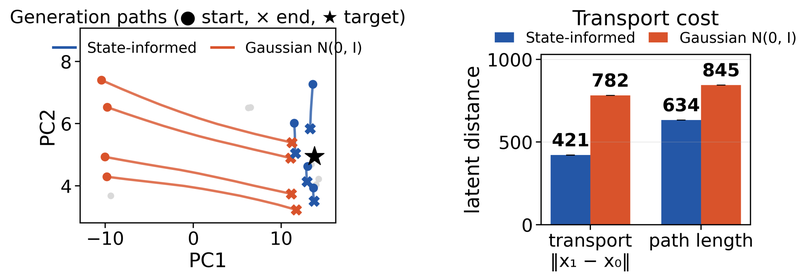
6 Ablation: Mass Error (lower is better)



Continuity operator and CorrectionNet are jointly necessary: removing either raises mass error > 10× (0.0114 → 0.1427 / 0.1319); the corrector absorbs discretisation error and keeps MAE (0.0093 → 0.0154 without it).

Decoder fine-tune: 0.0260 → 0.0114; **State-Informed Prior**: 0.0136 → 0.0114 (−16.2%).

7 Why the State-Informed Prior Helps



Additional analysis (not in the paper): one representative test case, 8 sampled paths, latent PCA, same target ★. Gaussian sources (red) start far away and travel long paths; state-informed sources (blue) start next to the target.

Anchoring at the observed state ⇒ **shorter transport** (cf. I²SB, DDBM for images).

Ablation agrees: the prior cuts mass error 0.0136 → 0.0114 (−16.2%) and improves calibration (Cal-50% 0.39 → 0.42) at zero MAE cost (paper, Table 1).

8 Key Takeaways

1. **Flow Matching for urban crowd-flow generation**: a probabilistic generator with physics inside the architecture
2. **Mass conservation as architecture**, not loss: −55.0% mass error with best MAE, RMSE and CRPS
3. **State-Informed Prior**: −16.2% mass error (0.0136 → 0.0114), better calibration, same MAE

Next: balance neural correction vs. hard physics; more cities and conservation-aware baselines; out-of-distribution events (large events, evacuation).

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References [1] Lipman et al. (2023). Flow Matching for Generative Modeling. *ICLR.* [2] Tong et al. (2023). Conditional Flow Matching. *arXiv:2302.00482.*
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